

7 **(i)** Express $\frac{1}{2r-1} - \frac{1}{2r+1}$ as a single fraction. **[2]**

(ii) Find how many terms of the series

$$\frac{2}{1 \times 3} + \frac{2}{3 \times 5} + \frac{2}{5 \times 7} + \dots + \frac{2}{(2r-1)(2r+1)} + \dots$$

are needed for the sum to exceed 0.999 999. **[7]**